(a) The learner changes visitation sensitivity

sensitivity $\partial E\theta/\partial \pi_s$

10^{2}
10^{1}
10^{0}
10^{-1}

10^{-3} 10^{-2} 10^{-1}

visitation d(s)

SGD: 1.00 (pred. 1)
Adam frequent: 0.49 (pred. 0.5)
Adam isolated: 0.97 (pred. 1)
Adam short, bias-corrected: 0.92 (pred. 1)
clipped SGD: 1.00 (pred. 1)

(b) Realized allocation error

realized mean absolute error

10^{-1}

0 4 8 12

allocation profile

water-filling
uniform
proportional to d
SGD weights

(c) Allocation gains: prediction vs realization

realized MAE ratio to water-filling

1.00 1.25 1.50 1.75

predicted MAE ratio to water-filling

y = x
uniform
proportional to d
SGD weights

(d) Scale invariance separates learners

mean absolute error

10^{0}
10^{-1}
10^{-2}
10^{-3}

10^{0} 10^{1} 10^{2} 10^{3} 10^{4} 10^{5}

demonstrations K

always-clipped SGD
SGD (worst of two worlds)
SGD aliasing floor
clipped RMS upper bound

Stored state-owned-coordinate experiments; sensitivity slopes describe the specified regimes.